

Chromosome Genetics

*Niels Larsen, PhD.
Dept. of Biochemistry*

Please review the semester 1 chromosome
lecture before this class

It is still true for all my lectures: what I present
orally is also part of what you will be tested on

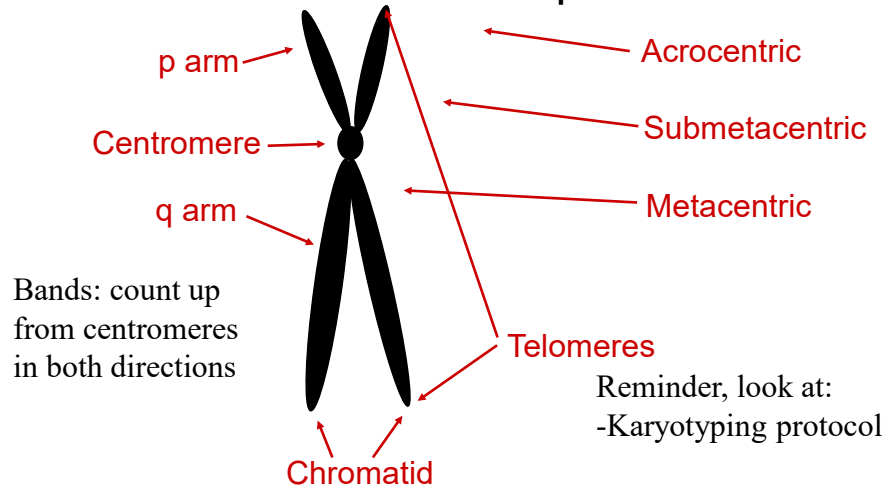
Learning objectives for the chromosome lecture in Reproductive module II

- Review chromosome structure and nomenclature of chromosome aberrations
- Look at telomere structure in more details: why does the telomere have the structure it does?
- Trisomy mechanism including interpreting quantitative marker analysis to determine the origin of extra chromosome
- Describe Mosaicism, including mechanism and application to all applicable examples
- Be prepared to answer first or second order vignette questions for selected chromosome disorders:
 - Turner syndrome, including symptoms, cause, effect on fetal survival, treatment
 - Klinefelter syndrome: symptoms, cause, physicians role
 - XXX, XYY: symptoms
 - Down syndrome and review of Robertsonian translocations
 - Edwards and Patau syndromes: symptoms, causes
 - Cri du Chat syndrome: symptoms, cause, variability, possibility of translocation carrier status in a parent
 - DiGeorge syndrome: symptoms, cause, *FISH* (fluorescent *in situ* hybridization) for diagnosis
- Understand the mechanism of FISH, chromosome painting, and comparative genome hybridization; explain the advantages and disadvantages of each and compare to regular banded karyotype
- Review indications for karyotyping
- Sample questions will be available in Canvas
- Reading: Emery's elements of medical genetics 15th ed. chapter 17 to page 241

Part 1

Slides with review material for your convenience – these will not be taught in class but are expected background knowledge from previous lectures

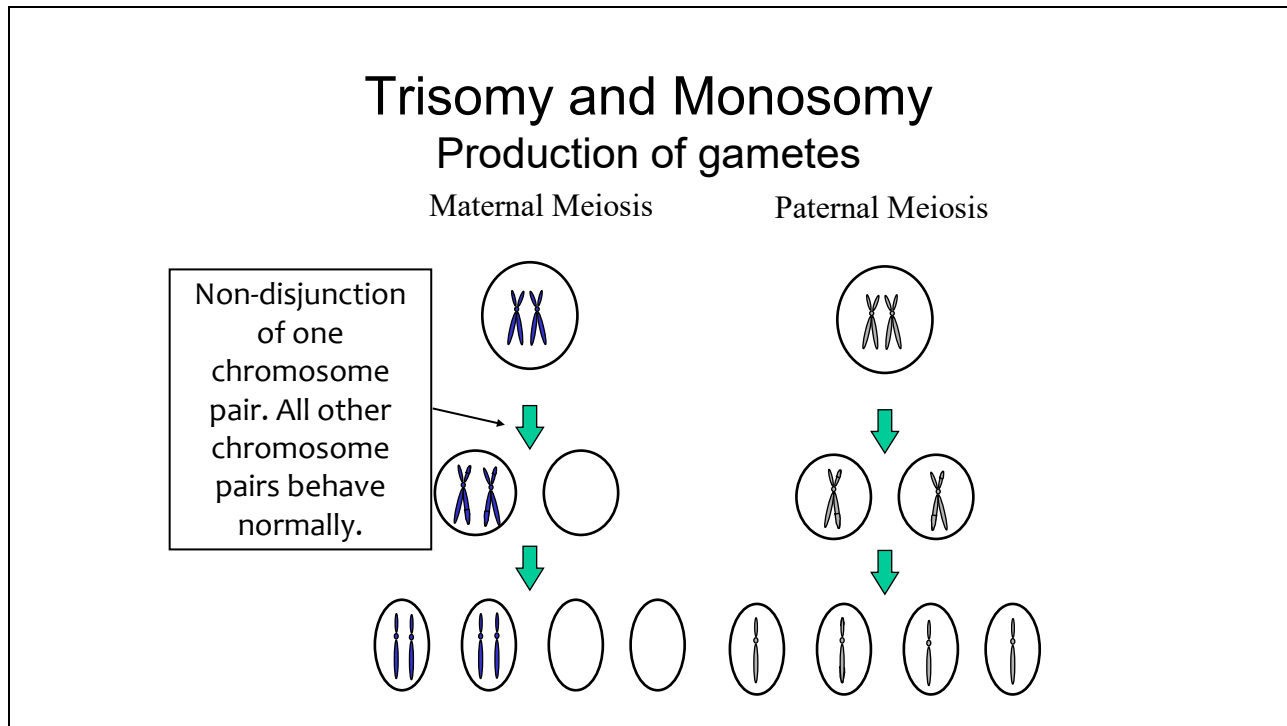
The Chromosome Shown in Metaphase



Numerical Aberrations

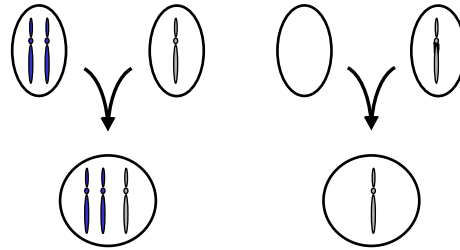
- “Wrong number” = aneuploid (opposite = euploid)
- Most commonly, that would amount to:
- **Polyploidy**: 3 (rarely 4) copies of all chromosomes; not viable; 3 copies = triploid (No = 69)
 - One of most common reasons for early pregnancy loss
- **Trisomy**, excess of one chromosome (e.g., 47, XX, +21)
 - Only trisomy of 13, 18, 21, or X,Y are compatible with life
 - Trisomy 16 is the most common of all
- **Monosomy**, lack of one chromosome
 - Only monosomy X is compatible with life
- **Causes**: non-disjunction or premature separation of chromatids

It is pretty important you know these definitions, including such as: what is the difference between trisomy and triploidy



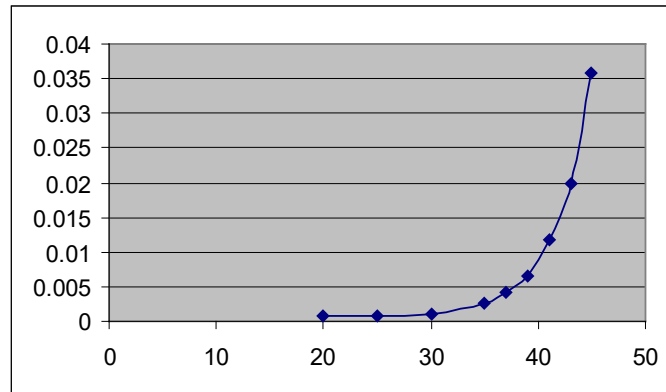
The error of meiosis here shows as first maternal meiotic division which is the most frequent location of the error causing Down syndrome. The other meiotic divisions can also cause monosomy or trisomy.

Trisomy and Monosomy (fertilization)



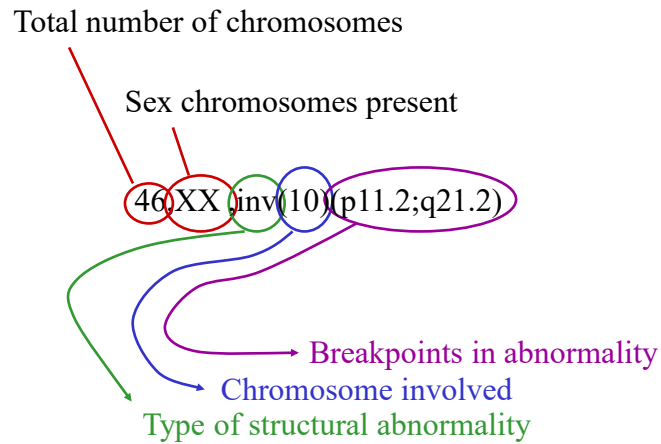
- In trisomy 21, 80 % of conceptions die *in utero*
- Monosomy is only viable with X-chromosome (more than 99 % of these die *in utero*)

Maternal Age Effect



Down's syndrome as function of maternal age
only found in diseases caused by nondisjunction, not by translocations.

Karyotype Nomenclature



You need to be able to read a karyotype written like this, and you should be able to distinguish balanced from unbalanced presentations

Types of Structural Abnormalities

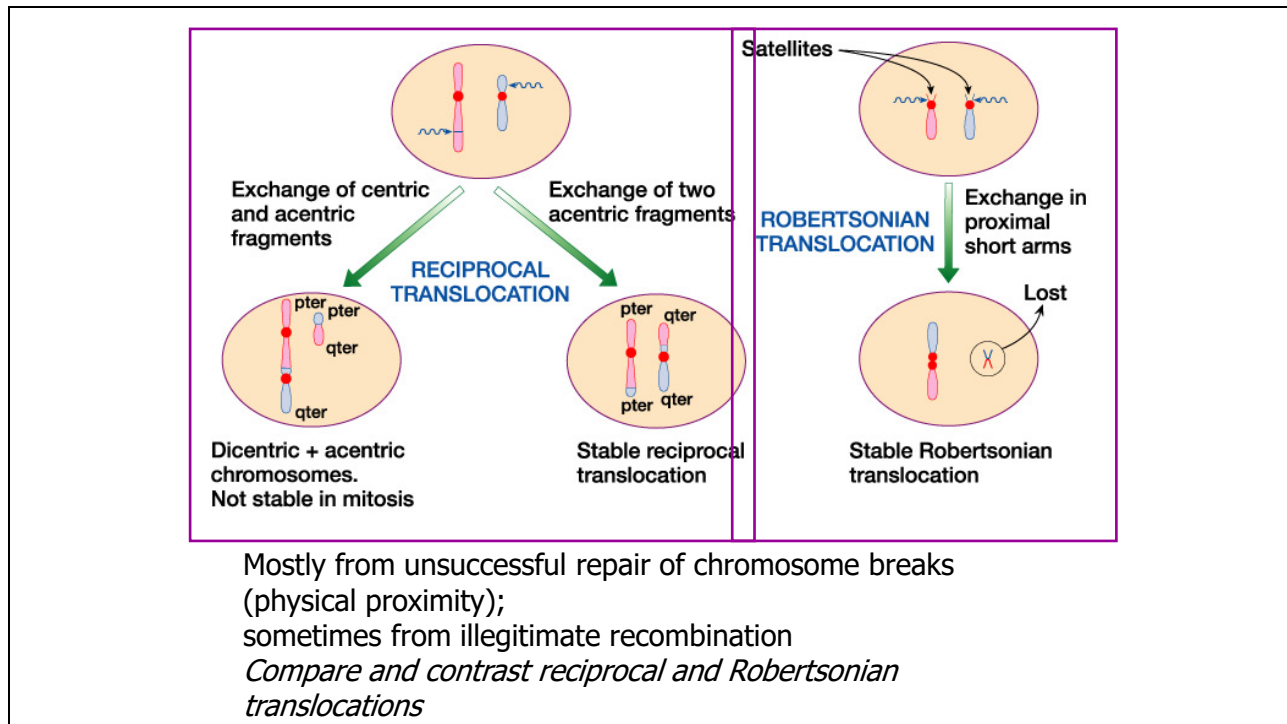
- Deletion (del)
- Inversion (inv)
- Duplication (dup) [a segment is present where it usually should be and in an additional position – usually next to the original]
- Insertion (ins) [a segment duplicated onto a different chromosome]
- Ring (r)
- Translocation, reciprocal (t)
- Translocation, Robertsonian (der) or (rob)
- Isochromosome (iso)

For this and the following slides: You should be able to deduce which of these is described in a verbal description, including knowing whether a translocation is reciprocal or Robertsonian.

Extra explanation for a couple of these:

Duplication: presence of two copies of a region (sometimes) on the same chromosome, and it is still present on the other copy of the chromosome so **three** copies total.

Insertion: presence of an extra chromosome piece coming from somewhere else in the genome where it is no longer present, so **two** copies of the area in total.



Notice that reciprocal translocations can involve any chromosome and that the breakpoints usually are in chromosomal arms; Robertsonian translocations only involve acrocentric chromosomes and the breakpoints result in a centromere fusion of the chromosomes.

02_21.jpg from Human Molecular Genetics

Chromosomes in Cancer

- In a few cases, the primary defect seems to be a chromosome defect
 - Most often *de novo* translocation
 - CML, ALL, etc
- Many more types of cancer show unstable chromosomes during later progression
 - Non-disjunction in mitosis
 - Structural changes

As Cancers Progress

- Accumulation of mutations that allow faster cell divisions etc
- Visible as
 - Aneuploidy (incorrect number of chromosomes)
 - Homogeneously staining regions
 - Double minutes
 - Marker chromosomes
- Often have prognostic value

Double minutes (DM)

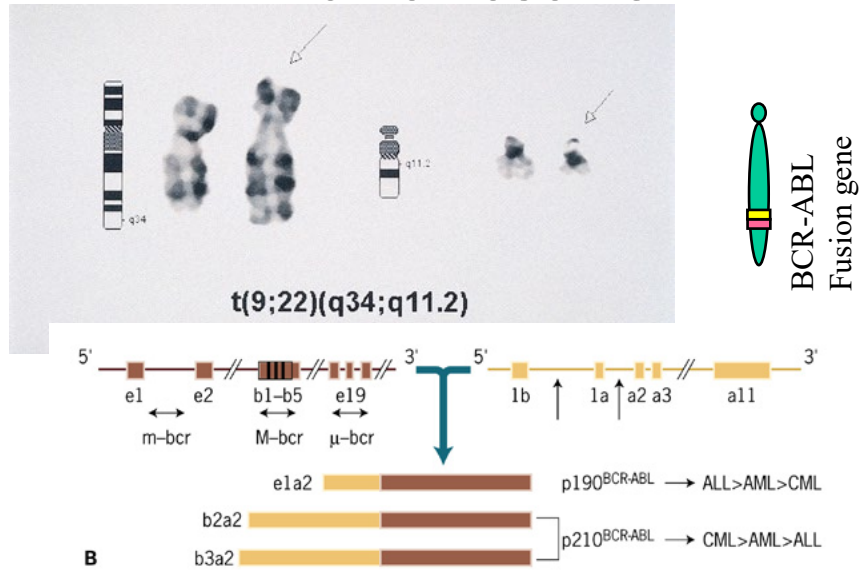
- Receptor tyrosine kinases often involved
- Order of events: gene amplification first, gene mutations second
- Somewhere along that route, excision from chromosome background allow more amplification (called homogeneously staining region prior to that)
- Do not contain centromeres, so segregation random – in situ selection for cells containing high content of mutated DM
- In cell culture often loss of double minutes!

Abnormal chromosomes in a cancer cell line



Result: 54, XY, +X, add(1)(q34), +4, +6, -7, +9, +10, +14, +18, +3mar

Ph chromosome



Chronic Myelogenous Leukemia (CML)

- The BCR-ABL fusion gene is the hallmark of CML
- Normally presents in middle age
- Involves overproduction of white-blood-cells
- Patients initially notices fatigue, night sweats and low grade fever
- Examination shows enlarged spleen and sometimes sternum tenderness

CML (continued)

- The initial phase can last for month to years
- Then accelerated phase: fever, bone pain, and splenomegaly
- Last phase: “blast-crisis”, bleeding and infections related to bone marrow failure

Diagnosis

- physical findings raise suspicion
- decreased leukocyte alkaline phosphatase indicative
- The Philadelphia chromosome and/or BCR-ABL fusion gene is conclusive
- BCR-ABL in most white blood cells already in the chronic state

CML (continued)

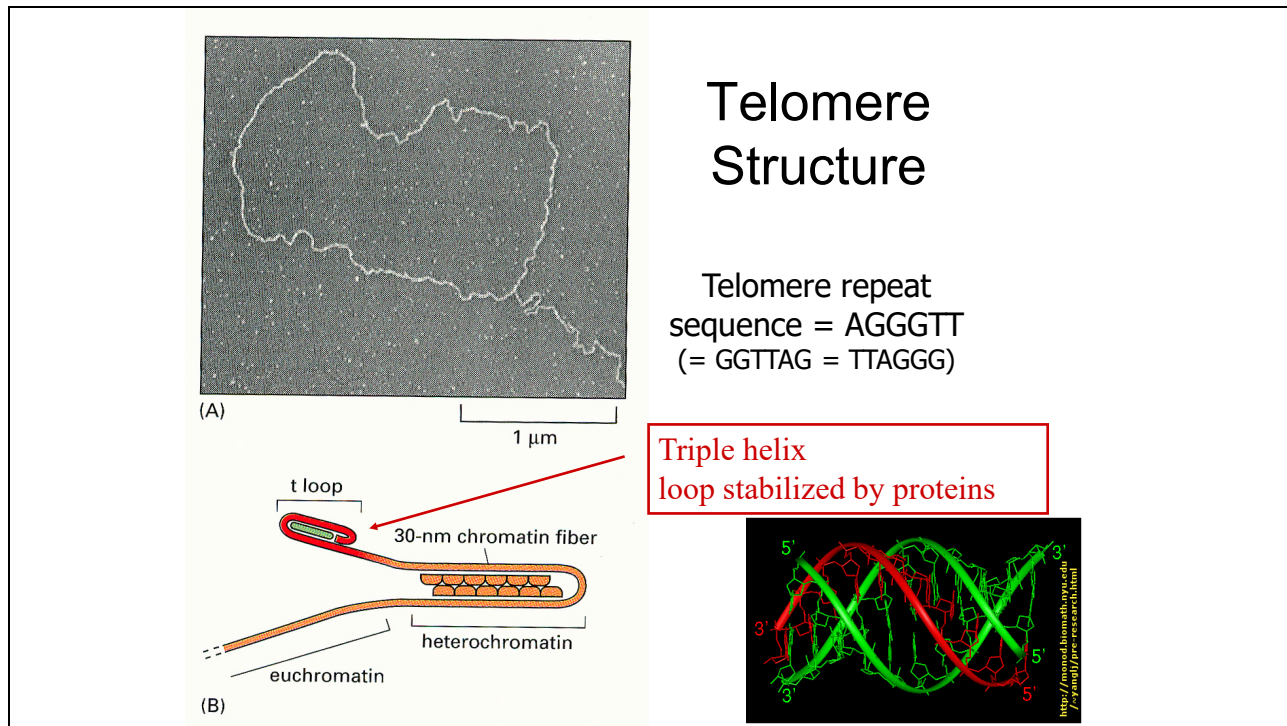
- Patients invariably die from the disease if left untreated
- Since 2001, tyrosine kinase inhibitors seems to control the disease in early stages and to cause improvement even in advanced cases
- Bone marrow transplantation can cure the patient if a matched donor can be found
- Hydroxyurea ingestion can alleviate symptoms
 - Inhibition of dNTP synthesis

Tyrosine Kinase inhibitors

- Imatinib mesylate was the first
- This drug is designed as a drug that inhibits the BCR-ABL activity
- It is currently the first response option for early phase patients (some patients have been relapse free for 5-6 years)
- Bone marrow transplantation is still the only treatment known to completely cure the disease
- (Brand names Gleevec or Glivec; second generation drugs are being used to some extent)

Part 2

Slides with new materials

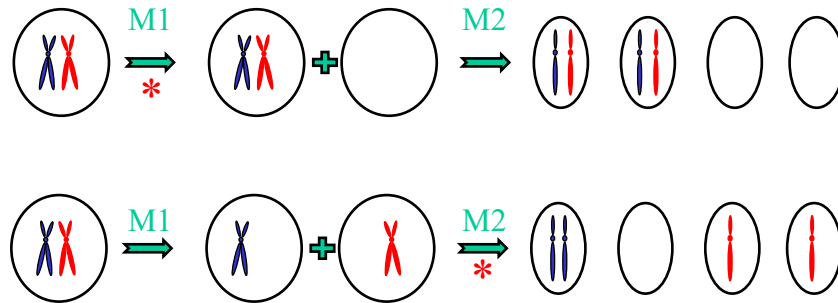


The loop structure stabilized by the triple helix and by interaction with proteins protect the end of the chromosome from break down

Telomere Structure

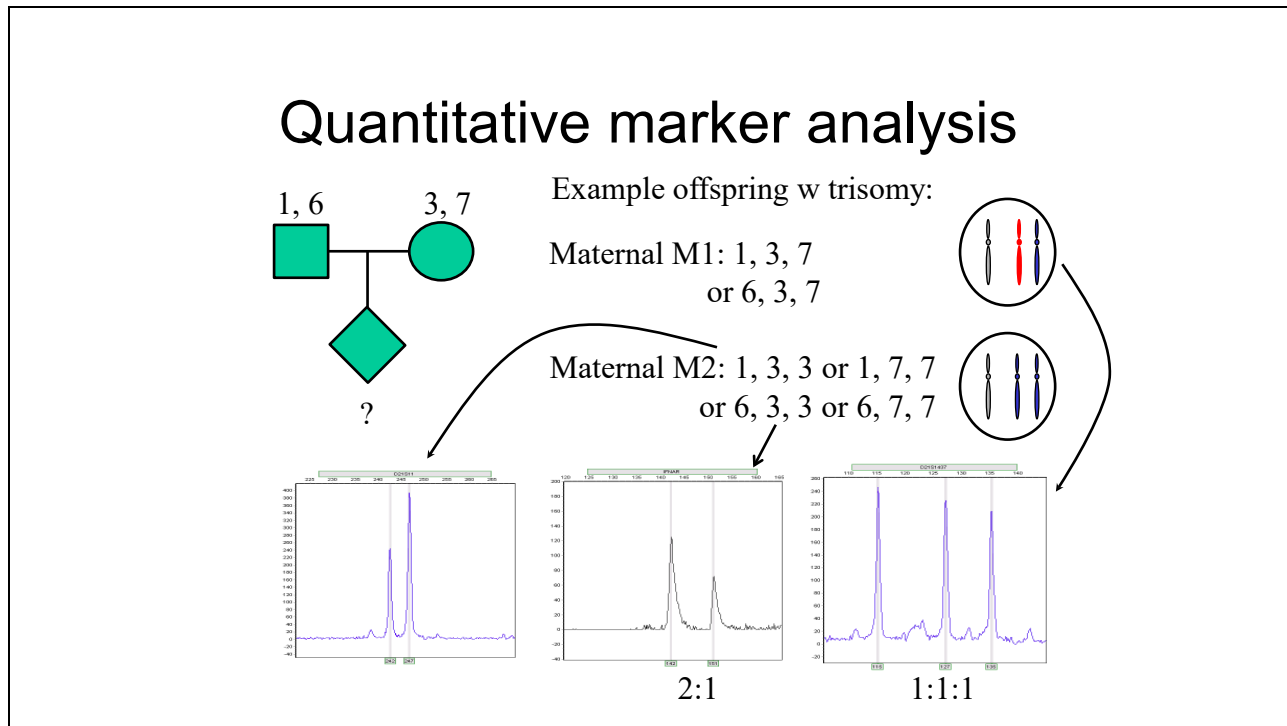
- Always have one repeat unit that is single stranded at the tip of the 3' strand
- This takes the form of a foldback triple-strand structure
- This structure is recognized by special proteins that bind to and protect the end from degradation

Effect of errors in 1st vs 2nd meiotic division



The take home message here is that the error in meiosis 1 would result in gametes having two **different** copies of the chromosome, while the error in meiosis 2 results in two **identical** copies of the chromosome going into the gamete. (Obviously, not taking the possibility for cross-over into account).

This conclusion will then be applied on next slide.

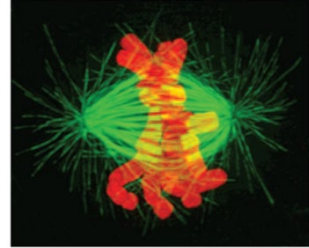


Figures from <http://www.softgenetics.com/Trisomyapplicationnote.pdf>

The right hand result shows three different alleles indicating that three different copies of a chromosome is found in that zygote. Logically, that means one gamete must have contained two different copies → error in meiosis 1. The parent in which the error happened can be seen from the marker analysis: if both allele 3 and 7 are present then the mother gave 2 different alleles and the meiosis 1 error happened in her.

The two results at the left shows one smaller peak and one that contains twice the material. Therefore, one gamete must have contained 2 ***identical*** copies of the chromosome. If the allele present in double amount is 3 or 7, then that error in meiosis 2 must have happened in the mother.

Cells for Karyotyping: Metaphase



- Adult
 - Blood (leukocytes): culture 2-3 days
 - Bone marrow biopsy: no culture
 - Fibroblast cells: culture 2-3 weeks
- Prenatal
 - Amniotic cells: culture 2-3 weeks
 - Chorionic villus sample: culture not necessary, but separation from maternally derived tissue **is** necessary

White blood cells is the workhorse of karyotyping. More details will be given about the two prenatal types in the lecture on Maternal serum screening.

Case Story

- A 14-year-old girl presents with the concern that she has not begun to menstruate. On physical examination, she has no breast development, but she does have Tanner IV pubic hair. She is the shortest child in her 8th grade class, but has grown 2 inches since last year



Turner Syndrome

Diagnosis

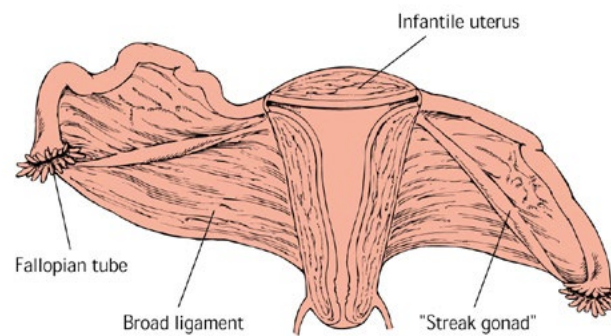
- Short stature, undeveloped/degenerated ovaries, missing development of secondary sexual characteristics
- Most are broad, have widely spaced nipples
- Normal intelligence
- Probably had a skin fold at the neck at birth (“webbed”)



The webbed neck is a result of lymphedema which is caused by reduced or delayed development of lymphoid system. Another frequent result of this is edema in hands and feet of infants with Turner.

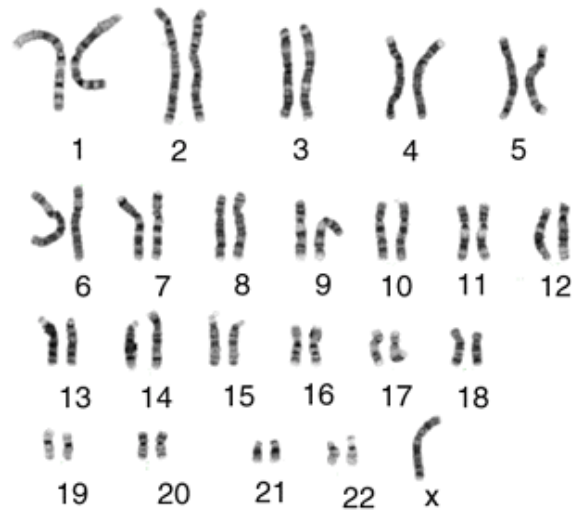


Above: 2-yo Turner patient
Right: internal repr. organs in adult



Hand shape frequently seen in Turner S; webbed neck; drawing of reproductive organs from adult with Turner S.

Karyotype in Turner Syndrome



Monosomy X, 45X

Turner Syndrome (3)

- Complications
 - Heart defect (elongated aortic arch, bicuspid aortic valve, aortic dilation, and coarctation of the aorta especially common), renal defects (30%), osteoporosis, and more rarely other (including isolated hypertension)
- Conclusion: relatively mild symptoms after birth
- Before birth, more than 99% of Turner conceptions lead to spontaneous abortion
 - Ultrasound findings: Nuchal translucency, cystic hygroma, sometimes heart defects or renal defects

Reminder: complications are conditions found in less than 50% of patients with a given diagnosis. The cardiac defects were also taught in the CARD module.

Nuchal Translucency

- Ultrasound of fetus
- Space between skin and soft tissue at spine
- Abnormally large NT indicates abnormality
- Invisible nasal bone indicates abnormality
- Several other possible outcomes than Turner:



Including normal!

More about this in the lecture on Maternal serum screening.

Turner Syndrome

- Incidence of 1/2500 (to 1/5000)
- Monosomy X in 40-50%
- Mosaic (46XX/45X or 46XY/45X) in 30-40 % of patients
- A number of other rarer combinations, many in mosaicism
- 46,X,i(Xq) (isochromosome for long arm of X) also presents with Turner syndrome (rare)
- Turner syndrome: 80% have the maternal X present, pointing to a possible defect in the paternal meiosis and/or a post-conception/mitotic error

A person with mosaicism is likely surviving better to birth than a person with all cells showing the monosomy. (Remember that 99% of conceptions with monosomy X do not make it to live birth.)

That a person with isochromosome Xq will also present with Turner syndrome is one of those observations that (at least in the past) would be jumped on by question writers.

Regarding the frequency of mosaicism: UpToDate used to state 30-40% but has recently changed that to approximately 50% of patients have mosaicism.

Turner Syndrome “Treatments”

- Estrogen is normally given from about 12 y.o.a. to induce secondary sexual characteristics
- Growth hormone from age 8 to increase height (from below 5 feet; increase with about 3”) is common but controversial
- Check for virilization regularly: if present indicates “hidden” Y-chromosome material
 - If present => removal of gonads due to risk for malignancies
- Follow up on any complication

Treatment is in quotation marks because there is no treatment that will change the chromosome content of the cells, no treatment will cure the underlying condition. The items included here will change the phenotype somewhat.

I expect that Dept. Pharmacology will tell you about these treatments in much more detail than it is done here.

The removal of gonads would obviously be a recommendation to the patient who always should make the final decision.

Mosaicism Revisited

- A mosaic has two cells that differ in content or expression
 - Origin is post zygote, => a mitotic error
- Germline mosaicism occurs when the gonads contain two groups of cells e.g. 46XY/47XXY or 46XX/46XX,t(3q:5p) BUT rest of body is normal
- Germline mosaicism can lead to repeated occurrence of affected offspring
 - Risk of recurrence is therefore 1-2%, not 0% when the abnormality has not been found in parents
- Chimerism is presence of cells originating from two different zygotes

Not directly mentioned on the slide: a person with mosaicism will have a mixture of normal and abnormal cells. The presence of these normal cells will be expected to lead to less severe symptoms relative to the person who has only abnormal cells.

In germline mosaicism, it is impossible to detect abnormal cells in the blood and other body cells, only germline cells have the abnormality.

Klinefelter Syndrome (47,XXY)



- Occurs in 1/1000 males (increasing?)
- Boys with developmental delay and/or learning disabilities and social maladjustment
- **Or** men seeking help for infertility
 - One study: 77% had not been detected at the age of 40
- More than average length, long limbs, small testis producing reduced levels of testosterone, reduced sexual function
- 55 % experience gynecomastia
 - When breast development occurs, risk of breast cancer

Incidence in the British biobank: 213 XXY among 207,067 men aged 40-70 (1.03/1000). The paper describing this set of results stated that 23% of those with XXY had selfreported this condition or had a note of chromosomal abnormality in the health database that they had consented to being used in the study. <https://doi.org/10.1016/j.gim.2022.05.011>

Causes

- 56% extra maternal X
- 44% extra paternal X
- 15% have mosaicism

Two first lines are about source, and line three is what can happen afterwards no matter which parent gave the extra X.

Klinefelter's Cont'd

- Intelligence range from normal to intellectual disability, with an average of slight reduction in IQ
- Other complications present in some patients:
- Obesity, diabetes, problems in thyroid function, pulmonary disease, cardiovascular disease

Klinefelter's Cont'd

- Small testis -> low testosterone
- Through lack of feedback inhibition: high pituitary gonadotrophins
- Treatment with androgens
 - Improve virilization and bone density
 - May worsen gynecomastia
 - Does not fix lack of fertility

XXX

- 1:1000 females (47,XXX)
- Tall
- Some reduction in IQ among known cases
- 62% = no physical abnormality; → lots not diagnosed
- XXX have increased risk of intellectual disability, psychological disorders, seizures, premature ovarian failure, and a variety of other complications
- Maternal age effect
- Intellectual disability in 48,XXXX or 49,XXXXX
- Additional Barr bodies

<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2987225/>

<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC1946229/pdf/canmedaj01326-0003.pdf>

XYY

- 1:1000 males
 - British biobank study: 0.7/1000, 99% not detected before
- Increased height, decreased IQ (average 10-15 points)
- Rates of obesity, diabetes, and cardiovascular problems higher than 46XY

The British biobank study was the same that were mentioned in the notes for Klinefelter. Only 1 of 143 people with XYY had been detected prior to the study. Study participants were between age 40-70.

So: people with two sex chromosomes have normal height, those with one are short and those with three are tall.



Downs Syndrome

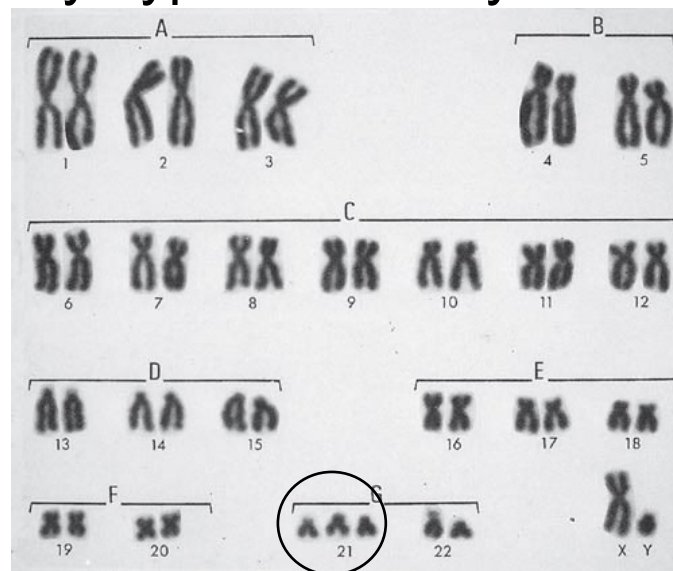


- Diagnostic signs include intellectual disability and specific shapes of the head and often hand and feet
- Complications: 40 % have heart malformations
 - Several others incl. duodenal atresia in 1-5%
- Hypothyroidism
- Down syndrome associated Myeloid leukemia (ML-DS) in 5-30%; presents before age 4, similar to AML; excellent prognosis when treated.



Single palmar crease, low-set, small ears with deviation from normal shape. Upslanting palpebral fissures, flat nasal bridge, enlarged tongue
AML = acute myeloid leukemia

Karyotype Downs Syndrome

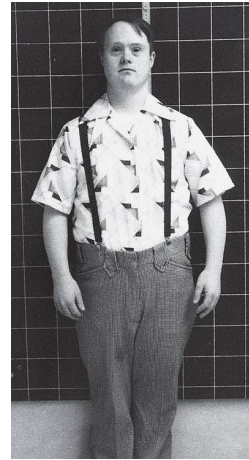
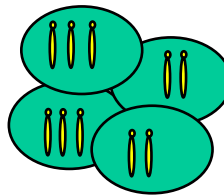


95% have this Karyotype

Causes of Downs Syndrome

- 95 % have classical trisomy 21 (47XX,+21 or 47XY,+21)
 - Most often an error in segregating the chromosomes in the first maternal meiosis
- 3-4 % have translocation (some partial reciprocal, most Robertsonian; 46XY,rob(14;21)),+21
- 1 % Mosaicism

Mosaicism caused by
non-disjunction in
mitosis



The increased risk with increased maternal age only apply to classical trisomy. The increased risk does not apply for parent carrying a Robertsonian translocation – but as you know from semester 1, that group does have a higher risk.

Downs Syndrome Continued

- Incidence at birth is 1/700 live births
 - Trend is for increasing incidence
 - First year survival about 90%
- However, the incidence at conception is much higher, it is estimated that more than 60% are spontaneously aborted and 20% are stillborn

The trend for increasing incidence is likely due to the trend of increased maternal age

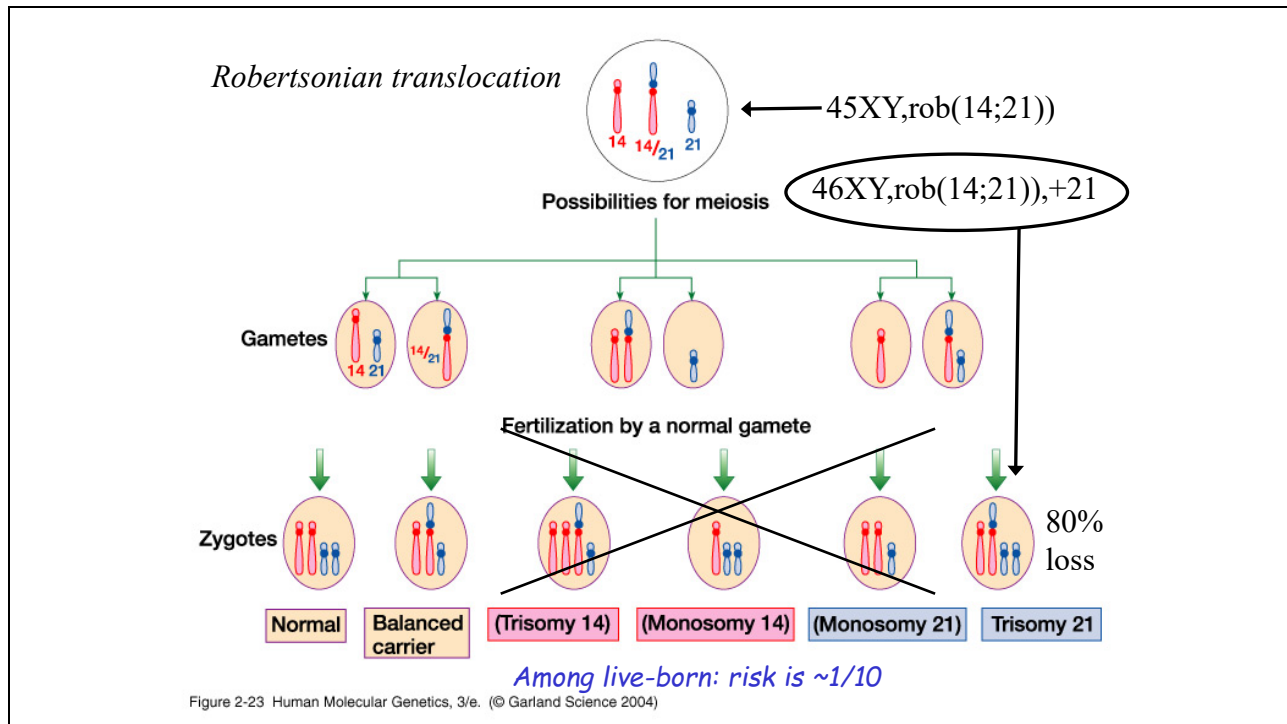
Ageing of Down Syndrome Patient

- Often experience early senility
- Some of this is due to Alzheimer's disease
 - Two genes on chromosome 21 that are suspected to participate in this: the APP gene and the cholesterol transporter ABCG1 gene
 - Some can be alleviated with proper nutrition

Robertsonian 14-21 Translocation Carrier



45 chromosomes total even though the karyotype is balanced



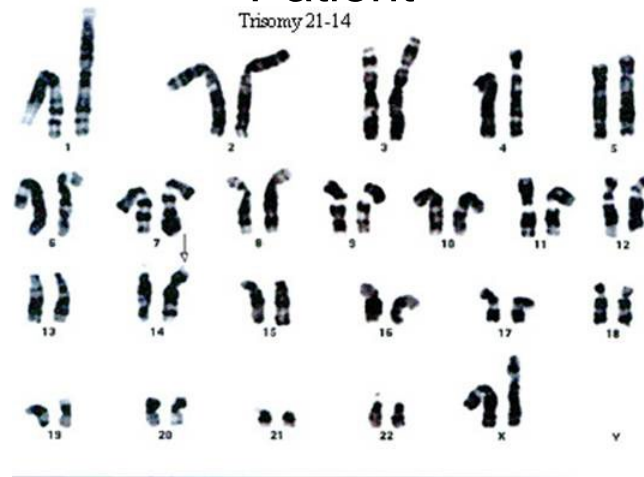
Two important outcomes from the slide not directly included in the slide:

- The family with one or more carriers of a balanced Robertsonian translocation has a possibility for several offspring affected by Down syndrome
- The carriers also have a risk of spontaneous abortions

If seeing those in a family, then Robertsonian translocation should be high on the list of differentials for explaining what is going on.

02_23.jpg

Robertsonian 14-21 Translocation Patient



46 chromosomes but unbalanced

Edwards Syndrome (47XX/XY,+18)

- Hypertonicity, clenched fingers
- Prominent occiput
- Low-set, malformed ears
- Small chin
- Rocker-bottom feet
- Heart defects
- Failure to thrive
- Profound intellectual disability
- Trisomy in 90% (mosaic and translocation rare)
- 1:8000, 80% female
- Maternal age effect
- Most die within a year



Omphalocele due to muscle weakness in the abdominal wall – 15% of patients have chromosome abnormality – poor prognosis. It is fairly common in Edwards syndrome

Patau Syndrome (47XX/XY,+13)

- Sloping forehead
- Forebrain defects
- Eye abnormalities (most)
- Cleft lip and palate (most)
- Low-set, malformed ears
- Polydactyly
- Heart defects (88%)
- Profound intellectual disability
- Deafness
- Trisomy 13 (some translocations and mosaics)
- 1:20,000 (sex equal)
- Maternal age effect
- Most die within first year

Structural Abnormality

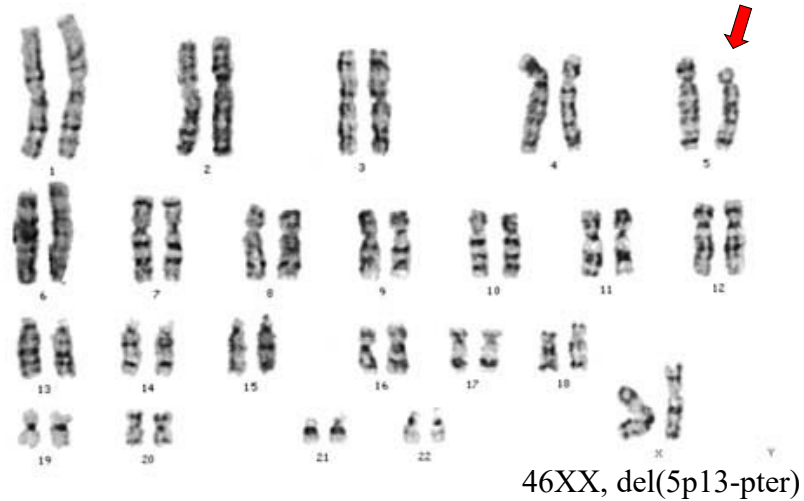


Cri-du-Chat Syndrome

- Symptoms
 - Very distinctive cry in babies
 - Low birth weight, failure to thrive, hypotonia, psychomotor retardation, microcephaly, hypertelorism, round face, antimongoloid slant of eyes, broad nasal bridge, and low-set and/or malformed ears
 - Most have severe intellectual disability, some only mildly
 - Characteristic facies



Cri-du-Chat Karyotype



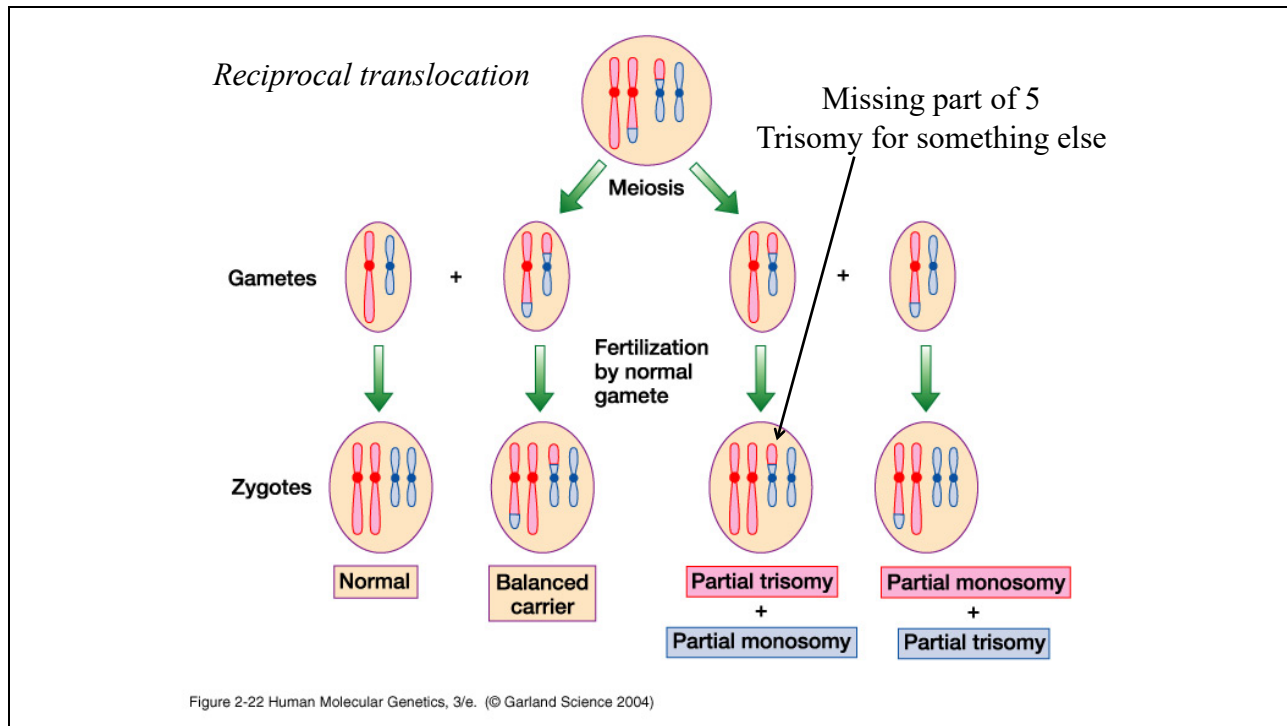
Cri-du-Chat Syndrome (3)

- Incidence estimates 1/20,000 to 1/50,000
 - Among patients w severe intellectual disability: 1/100 (1%)
- Deletions are variable in different patients
- Deletions include several genes (contiguous gene syndrome)
 - Larger deletion -> more intellectual disability
 - In very few cases, the cry can occur isolated, in patients showing none of the other symptoms

Cri-du-Chat Syndrome (4)

- Inheritance
 - 90% - spontaneous
 - 10% - one parent carries a **reciprocal** translocation
- Long term outlook?
- Italian Data:
 - Patients show good survival into their thirties
 - Poorer outlook for patients with translocation

If a parent carries a translocation, as you will see on next slide there is a risk for the next child as well. That also explains how a translocation can lead to a Cri-du-chat mimic



02_22.jpg

Reminder: overall, a parent carrying a balanced translocation has a $\frac{1}{2}$ risk of having a child with an abnormality. If the breakpoint is in the short arm of chromosome 5, symptoms similar to Cri-du-Chat will be in half of those children

In this slide, assume that the blue chromosome is Chr 5. Then you will see that the third zygote from the left is missing material of the distal part of the p-arm.

Results of Reciprocal Translocations

- Almost every case different, so few named syndromes; developmental defects common
- The unbalanced version affects multiple genes
- Leads to multiple affected tissues
- Common: intellectual disability and heart defects
- Anything is possible!
- Example karyotype balanced form: 46,XX, t(8;18)(p21.1;q21.1)

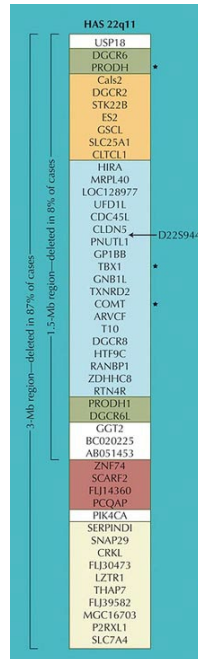
DiGeorge Syndrome/ CATCH 22/ VeloCardioFacial Syndrome

Reminder: Primary
Immunodeficiencies

- Structures from 3rd & 4th pharyngeal pouches under-/mis-developed
- Defects in aortic area of heart
- Hypoparathyroid -> hypocalcemia
- Reduced immune function (reduced or rarely absent thymus)
- Increased risk of psychiatric diseases from puberty
- **Highly variable phenotype!**
- Incidence 1/4000-1/6000

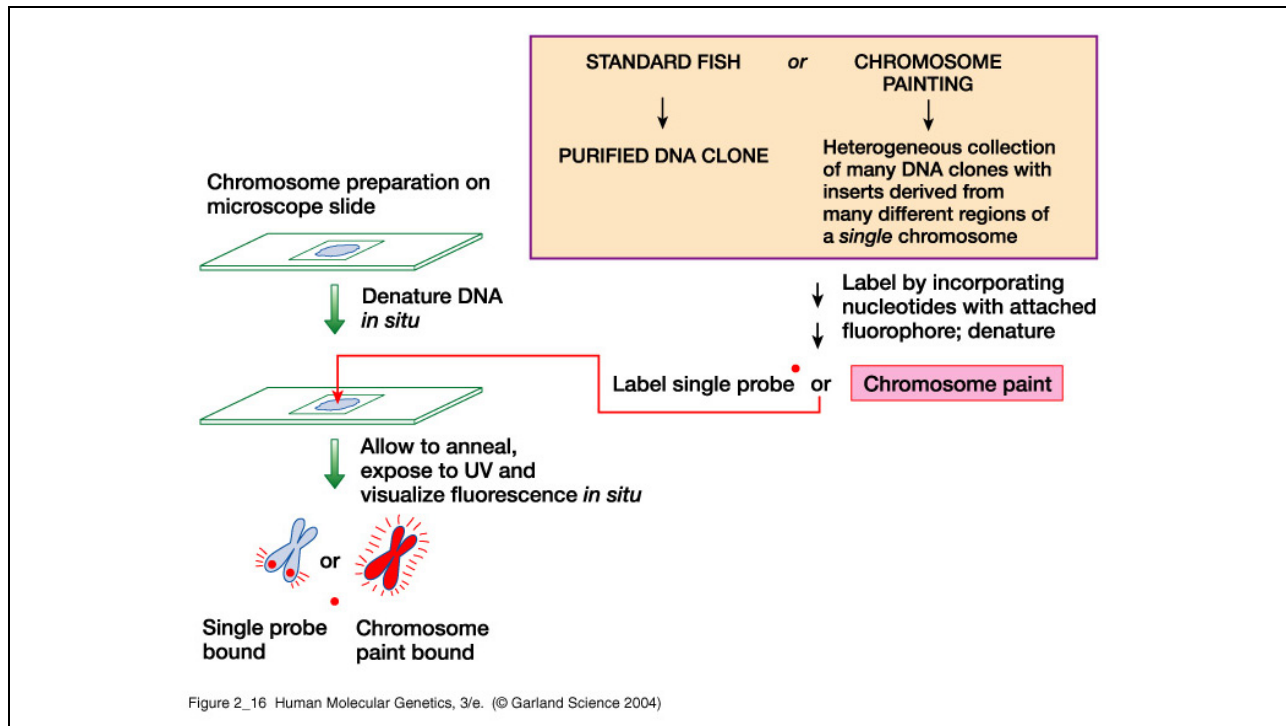


I realize you have heard about this condition elsewhere. I wanted to remind you of the disease and then use it to introduce the *FISH* procedure



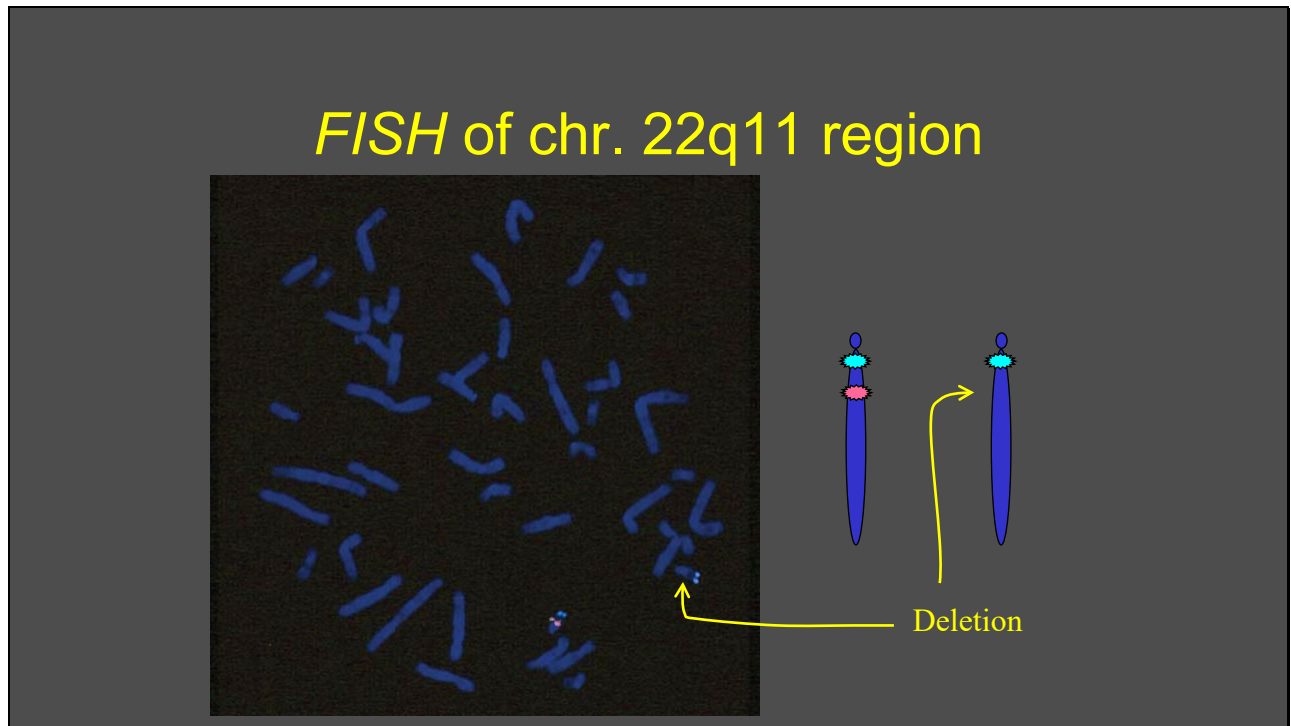
- Left: region of Chr. 22q11.2 deleted in DiGeorge S. >34 genes deleted in most cases
- 90% of cases show this area missing by *FISH*
- 2-5% deletion in 10p13-p14





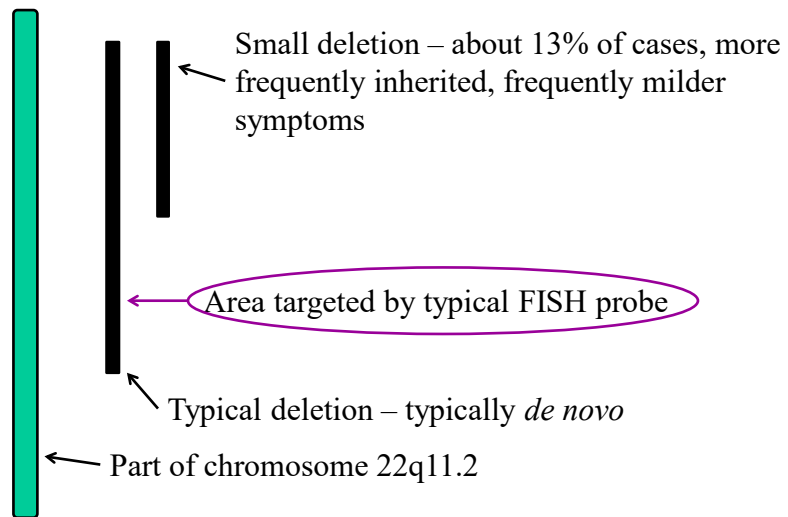
<http://www.nature.com/scitable/topicpage/fluorescence-in-situ-hybridization-fish-327>

02_16.jpg



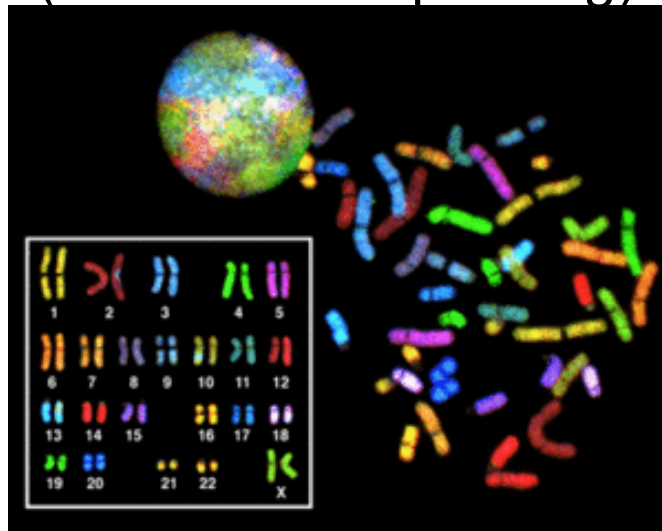
Notice that it is the absence of spots that indicate the deletion

Problem with DiGeorge region



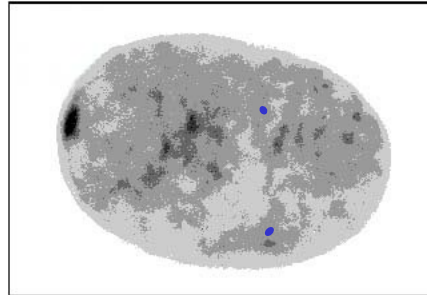
So the commonly used diagnostic tool is not ideal.

Spectral Karyotype (chromosome painting)

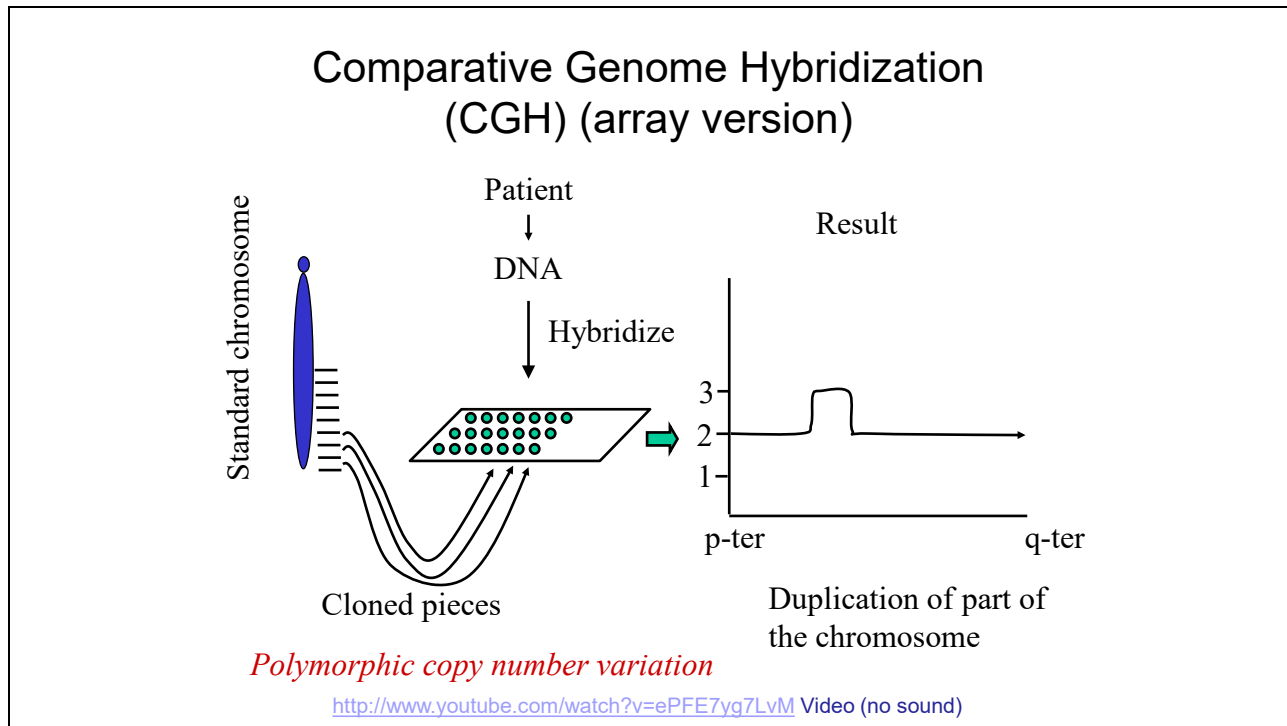


Spectral karyotype good for trisomies etc, great for translocations, not good for deletions, inversions, duplications within a chromosome, etc.

Interphase *FISH*



- Can be used to diagnose trisomies etc without waiting for cell growth
- Above image shows result of hybridizing a probe specific for one chromosome in normal cell
 - Three spots would be expected in case of trisomy



<http://www.youtube.com/watch?v=ePFE7yg7LvM>

The importance of polymorphic copy number variation is that deviation from normal content might not be pathological – in fact we all differ a little from each other.

Comparison FISH - CGH

- FISH requires: location known
- CGH: more complex and expensive, only use when FISH is not a good answer

Prenatal Diagnosis

- Use of CGH to analyze embryo biopsies finds more (up to 5% more) abnormalities compared to karyotype

Indications for Karyotyping

- Suspicion of recognized syndrome
- Unexplained intellectual disability
- Unexplained congenital malformations
- Pregnancy loss/infertility
- Abnormality in sexual development
- Malignancies